

NORMAL RELATIVELY CONVEX SUBGROUPS OF SOLVABLE ORDERABLE GROUPS

V. V. Bludov,^{1*} V. M. Kopytov,²
and A. H. Rhemtulla³

UDC 512.54

To the memory of N. Ya. Medvedev

Keywords: *ordered group, solvable group, convex subgroup.*

Orderable solvable groups in which every relatively convex subgroup is normal are studied. If such a class is subgroup closed then it is precisely the class of solvable orderable groups which are locally of finite (Mal'tsev) rank. A criterion for an orderable metabelian group to have every relatively convex subgroup normal is given. Examples of an orderable solvable group G of length three with periodic G/G' and of an orderable solvable group of length four with only one proper normal relatively convex subgroup are constructed.

INTRODUCTION

It is well known that every convex subgroup of an ordered locally nilpotent group is normal, and this is also true of ordered Engel groups; but this property does not extend to ordered solvable groups. It was shown in [1, Cor. 5.3.1] that every non-Abelian solvable ordered group has at least one proper (nontrivial) normal convex subgroup, and this is about the best that one can say in general.

In this paper, we study orderable solvable groups in which every relatively convex subgroup is normal. We show that if such a class is subgroup closed, then it coincides with the class of solvable orderable groups which are locally of finite (Mal'tsev) rank. This result is presented in Sec. 2. A criterion for an orderable metabelian group to have every relatively convex subgroup normal is given in Sec. 3. This is followed by examples of metabelian orderable groups having nonnormal convex subgroups for every order on the group. An example of a solvable orderable group G such that G/G' is periodic is taken up in Sec. 4, thus answering [2, Question 5.25]. A minor generalization of that example enables us to construct an orderable solvable group of length four with only one proper normal relatively convex subgroup (Example 3.4).

We give the notation and definitions. The sets of natural, integer, rational, and real numbers are denoted by $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$, respectively. We write $\gamma_n(G) = \underbrace{[G, \dots, G]}_n$ for the n th term of the lower central series

*Supported by RFBR (project No. 03-01-00320).

¹Irkutsk State Teachers Training University, Nizhnyaya nab. 6, Irkutsk, 664011 Russia; bludov@math.isu.ru.

²Sobolev Institute of Mathematics, Siberian Branch, Russian Academy of Sciences, pr. Akad. Kopt'yuga 4, Novosibirsk, 630090 Russia; kopytov@academ.org. ³Department of Mathematical and Statistical Sciences, University of Alberta, Edmonton, Alberta, T6G 2G1 Canada; ar@ualberta.ca.

Translated from *Algebra i Logika*, Vol. 48, No. 3, pp. 291-308, May-June, 2009. Original article submitted December 7, 2007; revised October 29, 2008.

in G and write $H \wr G$ for the restricted wreath product of groups H and G . A cyclic group of order p and an infinite cyclic group are denoted by C_p and C_∞ , respectively.

A subgroup H is *isolated* (*strictly isolated*) in a group G if $g^n \in H$ ($g^{1+f_1+\dots+f_n} \in H$) implies $g \in H$ for any $g, f_1, \dots, f_n \in G$, $n \geq 1$. If H is a subgroup of G then $I(H)$ denotes the *isolator* of H in G (i.e., the minimal isolated subgroup of G which contains H). We call G an R^* -group if $\{1\}$ is strictly isolated in G .

Definition 1. An $\mathcal{N}$ -group is an orderable group in which every relatively convex subgroup is normal. An $\mathcal{N}^s$ -group is one in which every subgroup is an $\mathcal{N}$ -group. By $\mathcal{N}$ and $\mathcal{N}^s$ we denote the classes of $\mathcal{N}$ - and $\mathcal{N}^s$ -groups, respectively.

Let $S(a)$ denote the semigroup generated by all conjugates of a in G .

Definition 2 [3]. We call G an R^+ -group if $\langle a^G \rangle / \langle b^G \rangle$ is periodic for every $b \in S(a)$.

1. $\mathcal{N}^s$ -GROUPS

It is well known that a wreath product $\langle a \rangle \wr \langle t \rangle$ of two infinite cyclic groups has nonnormal relatively convex subgroups (see [4; 5, Sec. 2.1]), and so $\langle a \rangle \wr \langle t \rangle$ is not an $\mathcal{N}$ -group. Therefore, an $\mathcal{N}^s$ -group does not contain a subgroup isomorphic to $\langle a \rangle \wr \langle t \rangle$. We also observe that a wreath product of two infinite cyclic groups $\langle a \rangle$ and $\langle t \rangle$ admits orders with both infinitely and finitely many normal convex subgroups. The infinite case is realized when a series of all convex subgroups is precisely a lower central series of a group. The finite case arises when we fix a positive integer n , require that $\gamma_i(\langle a \rangle \wr \langle t \rangle)$ are convex for $i = 1, \dots, n$, and then define the lexicographic order on $\gamma_n(\langle a \rangle \wr \langle t \rangle)$.

THEOREM A. A finitely generated solvable $\mathcal{N}^s$ -group has finite (Mal'tsev) rank.

LEMMA 1.1. Let G be a solvable orderable group that is locally of finite rank. Then every relatively convex subgroup is normal in G and every isolated normal subgroup of G is relatively convex.

Proof. If G has finite rank then every relatively convex subgroup of G is normal in G [4, Chap. II, Sec. 2, Prop. 5]. Suppose that G is locally of finite rank and let C be a relatively convex subgroup of G . Then $C \cap L$ is normal in L for every finitely generated subgroup L . If we take L to be the subgroup $\langle c, g \rangle$ we see that $c^g \in C$ for every $c \in C$ and every $g \in G$. Hence, C is normal in G .

Now let N be an isolated normal subgroup of G and suppose that G/N is not orderable. Then there is a finitely generated subgroup L of G such that LN/N is not orderable. Therefore, $N \cap L$ is not relatively convex in L . But every isolated normal subgroup of a finitely generated orderable solvable group of finite rank is relatively convex [5, Thm. 4.1.1]. Contradiction. $\square$

LEMMA 1.2. Suppose that G is a finitely generated solvable $\mathcal{N}^s$ -group such that G does not have finite rank but G' is locally of finite rank. Put $J = I(G')$. Then:

- (1) J is locally nilpotent;
- (2) the solvability length of J is $n - 1$, where n is the solvability length of G ;
- (3) there exist $t \in G \setminus J$ and $a \in J$ such that $H = \langle a, t \rangle$ has a quotient H/K that is isomorphic to $C_p \wr C_\infty$, with $\langle Ka \rangle$ the base group and $\langle Kt \rangle$ the top group.

Proof. (1) Let L be a finitely generated subgroup of G' and $\leq$ be an order on G . Let C be the union of all convex subgroups X_i of G such that $X_i \cap L < L$. Then $C \cap L < L$ since L is finitely generated. Let D be the intersection of all convex subgroups that contain L . Then $C \prec D$ is a jump and $[D, G'] \leq C$ because G' centralizes all convex jumps. Since $L \subseteq G'$, $[L, D] \leq C$, it follows that $L' \leq L_1$, where $L_1 = C \cap L$. We observe that L_1 is the normal closure in L of a finitely generated subgroup and that it is isolated in L .

Now let $C_1 \prec D_1$ be a jump, where C_1 is the union of all convex subgroups X_i of G satisfying $X_i \cap L_1 < L_1$, and D_1 is the intersection of all convex subgroups of G that contain L_1 . Then $[L, D_1] \leq C_1$, and hence $[L, L_1] \leq L_2$, where $L_2 = L_1 \cap C_1$. Note that L_2 is the normal closure in L of a finitely generated subgroup.

The process is repeated until we obtain a descending chain L_i of isolated subgroups of L . Since L has finite rank, $L_k = 1$ for some finite integer k , and hence L is nilpotent.

This shows that G' is locally nilpotent. Since J/G' is a periodic Abelian group, J is a locally nilpotent-by-finite orderable group, and so it is locally nilpotent.

(2) We use induction on the solvability length m of G' . If G' is Abelian, then so is J , since J/G is periodic and J is orderable. If $m > 1$, then we let A be a maximal normal Abelian subgroup of J containing the last nontrivial term of the derived series in G' . Then A is isolated in J and J/A is orderable by Lemma 1.1. The result now follows by induction since AG'/A is solvable of length $m - 1$.

(3) By [6], there are subgroups $K < H \leq G$ such that K is normal in H and H/K is isomorphic to $C_p \wr C_\infty$ for some prime p . Thus, we can take H to be the subgroup $\langle a, t \rangle$, in which case $H/K = \langle Ka \rangle \wr \langle Kt \rangle \cong C_p \wr C_\infty$. Now H is not contained in J for H is not nilpotent. More specifically, $t \notin J$ since its normal closure in H contains a subgroup $H_1 = \langle [a, t], t \rangle$ which has a quotient isomorphic to H/K . Replacing H with H_1 if necessary, we may assume that $a \in J$. This completes the proof of item (3). $\square$

LEMMA 1.3. Let $H = \langle a, t \rangle$ be a torsion-free metabelian subgroup of G with the following properties:

- (1) G has an isolated normal Abelian subgroup J such that G/J is Abelian;
- (2) there are elements $t \in G \setminus J$ and $a \in J$ such that $H = \langle a, t \rangle$, and for some normal subgroup K of H , $H/K = \langle Ka \rangle \wr \langle Kt \rangle \cong C_p \wr C_\infty$.

Then $H \cong C_\infty \wr C_\infty$.

Proof. Note that $K \subseteq B = \langle a^{(t)} \rangle$. Moreover, B is a direct product of cyclic groups $A_i = \langle a^{t^i} \rangle$ for B/K is a direct product of groups KA_i/K . Since B is torsion-free, the subgroups A_i are infinite cyclic, and $H = \langle a \rangle \wr \langle t \rangle \cong C_\infty \wr C_\infty$. $\square$

LEMMA 1.4. A finitely generated metabelian $\mathcal{N}^s$ -group has finite (Mal'tsev) rank.

Proof. We use the notation established in Lemma 1.2. The group J is Abelian since G is metabelian. By Lemma 1.2(3), there are $t \in G \setminus J$ and $a \in J$ such that $H = \langle a, t \rangle$ and $H/K = \langle Ka \rangle \wr \langle Kt \rangle$. Let $B = \langle a^{(t)} \rangle$. Then B is a direct product of infinite cyclic groups $A_i = \langle a^{t^i} \rangle$, and as shown in Lemma 1.3, $H = \langle a \rangle \wr \langle t \rangle \cong C_\infty \wr C_\infty$. Since a wreath product of two infinite cyclic groups is not an $\mathcal{N}$ -group, $G \notin \mathcal{N}^s$. $\square$

Proof of Theorem A. We use induction on the solvability length n of G . The case $n = 2$ was treated in Lemma 1.4. Further, we adhere to the notation created in Lemma 1.2. By induction, we may assume that G' , and hence (by Lemma 1.2(2)) J , will be locally of finite rank. By Lemma 1.2(1), J is locally nilpotent. Hence, J is solvable of length $n - 1$. Let J_i be the isolator of the $(n - 1 - i)$ th derived subgroup of J . Then $1 = J_0 \triangleleft J_1 \triangleleft \dots \triangleleft J_{(n-1)} = J$ is a torsion-free Abelian series and each J_i is normal in G .

If the rank of G is infinite, then it follows by [6] that G contains a 2-generator subgroup $H = \langle a, t \rangle$ having a quotient $H/K = \langle Ka \rangle \wr \langle Kt \rangle \cong C_p \wr C_\infty$. Let i be the least integer such that G has such a section with $a \in J_i$. If $i = 1$, then H is metabelian, and $H \notin \mathcal{N}$ by Lemma 1.4, a contradiction. We assume $i > 1$.

Every nontrivial normal subgroup of H/K contained in $K\langle a^{(t)} \rangle/K$ is an infinite direct product of cyclic groups of order p , and $\langle J_{i-1}, t \rangle$ is locally of finite rank by the choice of i . Hence, $B \cap J_{i-1} \leq K$, where $B = \langle a^{(t)} \rangle$. Since $H \cap J_{i-1}$ is locally nilpotent and the locally nilpotent radical of H/K is BK/K , it follows that $H \cap J_{i-1} \subseteq K(B \cap J_{i-1}) \subseteq K$. Thus, H/K is a quotient of $H/H \cap J_{i-1}$. Since $a \in J_i$, $B \subseteq J_i$. Moreover, H/B is infinite cyclic. Hence, $H/(H \cap J_{i-1})$ is a torsion-free metabelian group with isolated normal Abelian subgroup $B/(H \cap J_{i-1})$.

Let H be a finitely generated subgroup of an $\mathcal{N}^s$ -group G . Then H too is an $\mathcal{N}^s$ -group. Now we apply Lemma 1.3, with $H/(H \cap J_{i-1})$ replacing H and $B/(H \cap J_{i-1})$ replacing J in the statement of that lemma. It follows that $H/(H \cap J_{i-1})$ is isomorphic to a wreath product of two infinite cyclic groups. Such a group is orderable, and so therefore are H and $H/(H \cap J_{i-1})$. Hence, $(H \cap J_{i-1})$ is a relatively convex subgroup of H . But a wreath product of two infinite cyclic groups is not an $\mathcal{N}$ -group. Thus, there is an order on H with respect to which $(H \cap J_{i-1})$ is convex, and there are other nonnormal convex subgroups of H containing $(H \cap J_{i-1})$. This shows that $H \notin \mathcal{N}$, a contradiction. $\square$

PROPOSITION 1.5. If G is a solvable orderable group and all normal isolated subgroups of G are relatively convex then G is an R^+ -group.

Proof. Let $1 \neq a \in G$, $b \in S(a)$, $N(a) = \langle a \rangle^G$, and $N(b) = \langle b \rangle^G$. We prove the proposition by induction on the derived length of $N(b)$. If $N(b)$ is Abelian then its isolator $I(N(b))$ is Abelian and $I(N(b)) = \{g \in G \mid g^n \in N(b) \text{ for some positive } n\}$ (which is true for all nilpotent subgroups of orderable groups; see [5, Lemma 2.6.6]). In this case $I(N(b))/N(b)$ is periodic. By assumption, $I(N(b))$ is relatively convex and, hence, is strictly isolated. This gives $N(a) \leq I(N(b))$, and so $N(a)/N(b)$ is periodic as well.

Now we may assume that $N(b)$ is solvable of length $n > 1$. Let $A = (N(b))^{(n-1)}$ so that A is a normal Abelian subgroup of G and $N(b)/A$ is solvable of length $n - 1$. The isolator $I(A)$ is Abelian and is relatively convex in G . Consider an orderable group $\bar{G} = G/I(A)$. If $\bar{H}$ is a normal isolated subgroup of $\bar{G}$, then $I(A)H$ is isolated in G ; so $I(A)H$ is a normal isolated subgroup of G , and by assumption, $I(A)H$ is relatively convex in G . Hence, $\bar{H}$ is relatively convex in $\bar{G}$ and $\bar{G}$ satisfies the hypothesis of the proposition. By induction, $\bar{N}(a)/\bar{N}(b)$ is periodic, and so for every $g \in N(a)$, there exist a positive n and elements $d \in I(A)$ and $h \in N(b)$ such that $g^n = hd$. Since A is Abelian, there exists a positive m for which $d_1 = d^m \in A < N(b)$. Hence, $g^{nm} = h_1 d_1 \in N(b)$ (here h_1 is a product of conjugates of h , and so it belongs to $N(b)$). Thus, $N(a)/N(b)$ is periodic. $\square$

2. ORDERS UNDER WHICH ALL CONVEX SUBGROUPS ARE NORMAL

Every non-Abelian solvable ordered group has at least one proper (nontrivial) normal convex subgroup (see [1, Cor. 5.3.1]). In this section, we study the situation where all relatively convex subgroups are normal.

THEOREM B. Let G be an orderable metabelian group. If $\text{rank}(G/G') = 1$ then G admits a total order under which all convex subgroups are normal. If, in addition, G is finitely generated then G has a finite series of convex normal subgroups with Archimedean jumps.

Proof. Let A be an isolator of G' . It is known that A is Abelian, and it follows from the assumption of the theorem that G/A is torsion-free Abelian of rank 1. Therefore, G/A is isomorphic to an additive subgroup of rationals, and we can denote G/A by $T = \langle t^r \mid r \in B \rangle$, where B is a suitable subgroup of rationals. If $g_1, g_2 \in G \setminus A$ have equal images in G/A , then $a^{g_1} = a^{g_2}$ for all $a \in A$. Moreover, $\langle g \rangle \cap A = \{1\}$ for all $g \in G \setminus A$ because G/A is torsion-free. Without loss of generality, we may replace G with a split product $A \rtimes T$. Then we embed A into a divisible completion A^* , extend actions of T to A^* , and consider a split product $A^* \rtimes T$. Thus, let $G = A \rtimes T$ be an orderable metabelian group (i.e., a metabelian R^* -group), A be a divisible Abelian normal subgroup of G , and T be a torsion-free Abelian group of rank 1 written in the form $T = \langle t^r \mid r \in B \leq \mathbb{Q} \rangle$.

Choose an element $a \in A$. Let $H(a) = \text{SI}(\langle a \rangle^G) \leq A$ be a strictly isolated normal subgroup generated by a . The group $H(a)$ is relatively convex because $G/H(a)$ is a metabelian R^* -group. Now we establish

a total order on G under which $H(a)$ is convex and is ordered with an Archimedean order. Since $H(a)$ is relatively convex, we can find a total order on G under which $H(a)$ is convex, and if this order induces a non-Archimedean order on $H(a)$, then a and a^t are not Archimedean equivalent, i.e., $a \ll a^t$ or $a \gg a^t$. For all positive rational r , this gives $a \ll a^{t^r}$ in the first case and $a \gg a^{t^r}$ in the second. Therefore, for all $m_1, \dots, m_n \in \mathbb{Q}$ and all $r_1 < \dots < r_n \in \mathbb{Q}$,

$$a^{m_1 t^{r_1} + \dots + m_n t^{r_n}} = 1 \Rightarrow m_1, \dots, m_n = 0. \quad (1)$$

Any element $h \in \langle a \rangle^G$ has the form $h = a^{k_1 t^{r_1} + \dots + k_n t^{r_n}}$, $k_1, \dots, k_n \in \mathbb{Q}$, $r_1 < \dots < r_n \in \mathbb{Q}$, and so (1) holds for this h . Since $g \in H(a)$ satisfies the condition $g^{s_1 t^{r_1} + \dots + s_n t^{r_n}} \in \langle a \rangle^G$, (1) holds for all elements of $H(a)$. Thus, every element g of $H(a)$ is uniquely representable as $g = a^{\frac{f(t_1)}{h(t_1)}}$, where f and h are polynomials in $t_1 = t^{1/n}$ and n depends on g . Take a positive real τ and set $g > 1$ if $\frac{f(\tau^{1/n})}{h(\tau^{1/n})} > 0$, where f and h are taken from the representation of g . This gives us an Archimedean order on $H(a)$.

To complete the proof, we well order A , take the first element $1 \neq a_1 \in A$, and define a total order on G with Archimedean ordering on $H(a)$ and $H(a)$ convex. Let a total order be defined on G with convex $H(\{a_\alpha\}_{\alpha < \beta}) = \text{SI}(\langle a_\alpha \mid \alpha < \beta \rangle^G)$ such that all convex subgroups contained in $H(\{a_\alpha\}_{\alpha < \beta})$ are normal. If $a_\beta \in H(\{a_\alpha\}_{\alpha < \beta})$ then we put $H(\{a_\alpha\}_{\alpha \leq \beta}) = H(\{a_\alpha\}_{\alpha < \beta})$. If not, consider $\bar{G} = G/H(\{a_\alpha\}_{\alpha < \beta})$. $\bar{G}$ satisfies the hypotheses of the theorem, $\bar{a}_\beta \neq 1$, and so we can establish a total order on $\bar{G}$ with Archimedean ordering on $\bar{H}(a_\beta)$ and convex $\bar{H}(a_\beta)$. Let $H(\{a_\alpha\}_{\alpha \leq \beta})$ be a preimage of $\bar{H}(a_\beta)$ in G . By transfinite induction, we define on G a total order with a well-ordered series of convex subgroups of the form $H(\{a_\alpha\}_{\alpha < \beta})$ or $H(\{a_\alpha\}_{\alpha \leq \beta})$ and with the last subgroup A . All such groups are normal in G .

If G is a finitely generated, then G satisfies the maximal condition for normal subgroups, so any well-ordered ascendant chain of normal subgroups is finite, and hence we obtain an order on G with a finite series of normal convex subgroups with Archimedean jumps. $\square$

PROPOSITION 2.1. For every $n > 1$, there exist orderable metabelian groups G_n with n generators having nonnormal convex subgroups for any ordering of G_n .

Proof. It suffices to construct a 2-generator group with the required properties. Let $M = M(x, y)$ be a free metabelian group with free generators x and y . We put $c = [x, y]$ and denote by N a normal subgroup in M generated by $c^{x+(y-1)^2}$, i.e., $N = \langle c^{x+(y-1)^2} \rangle^M$. Let $G = M/N$.

To show that G is orderable, it is enough to verify that G is an R^* -group. Let $h \in G$ so that

$$h^{n_1 g_1 + \dots + n_k g_k} = 1 \quad (2)$$

for some $g_1, \dots, g_k \in G$ and for positive integers $n_1, \dots, n_k$. Since $N \leq M'$, G/G' is free Abelian, and hence $h \in G'$. In this case h is a product of conjugates of c and their inverses, i.e., $h = c^{f_1(x^{\pm 1}, y^{\pm 1})}$, where $f_1(x^{\pm 1}, y^{\pm 1})$ is a polynomial in $x^{\pm 1}, y^{\pm 1}$ with integer coefficients. For metabelian groups, any g_i in (2) can be taken to be of the form $g_i = x^{q_i} y^{s_i}$, $q_i, s_i \in \mathbb{Z}$, $i = 1, \dots, k$. In this way, (2) is rewritten thus:

$$c^{f_1(x^{\pm 1}, y^{\pm 1}) f_2(x^{\pm 1}, y^{\pm 1})} = 1, \quad (3)$$

where f_2 is a polynomial with strictly positive coefficients.

Thus, in the free metabelian group M , relation (3) gives us $c^{f_1(x^{\pm 1}, y^{\pm 1}) f_2(x^{\pm 1}, y^{\pm 1})} = c^{(x+(y-1)^2) f_3(x^{\pm 1}, y^{\pm 1})}$. Conjugating both sides by suitable powers of x and y yields the following relation in M :

$$c^{g_1(x, y) g_2(x, y)} = c^{(x+(y-1)^2) g_3(x, y)}, \quad (4)$$

where $g_1(x, y) = x^{t_1}y^{t_2}f_1(x^{\pm 1}, y^{\pm 1})$ and $g_2(x, y) = x^{t_3}y^{t_4}f_2(x^{\pm 1}, y^{\pm 1})$ (hence $g_2(x, y)$ is a polynomial with positive coefficients). Since M is free metabelian, relation (4) implies a polynomial relation $g_1(x, y)g_2(x, y) = (x + (y - 1)^2)g_3(x, y)$, and so $(x + (y - 1)^2)g_3(x, y)$ divides $g_1(x, y)g_2(x, y)$. Since $(x + (y - 1)^2)$ is irreducible, $(x + (y - 1)^2)$ divides either $g_1(x, y)$ or $g_2(x, y)$.

We claim that $(x + (y - 1)^2)$ cannot divide $g_2(x, y)$. Suppose the contrary, letting $(x + (y - 1)^2)g_4(x, y) = g_2(x, y)$. Write $g_2(x, y) = x^k(g_{2,0}(y) + g_{2,1}(y)x + \dots + g_{2,n}(y)x^n)$. Since $(x + (y - 1)^2)$ does not divide a polynomial x^k , we may assume that $g_{2,0}(y)$ is nonzero, say, $g_{2,0}(y) = m_0 + m_1y + \dots + m_ky^k$. Recall that all coefficients of $g_2(x, y)$ are positive, and hence $m_0 + m_1 + \dots + m_k \geq 1$.

Taking $y = 1 + \varepsilon$ in a small neighborhood of 1 and putting $x = -(y - 1)^2 = -\varepsilon^2$, we obtain $(x + (y - 1)^2) = 0$. On the other hand, $g_2(x, y)$ belongs to a small neighborhood of $g_{2,0}(y)$ which belongs to a small neighborhood of $m_0 + m_1 + \dots + m_k$; so $g_2(x, y)$ is not equal to zero, a contradiction. Thus, $(x + (y - 1)^2)$ divides $g_1(x, y)$; hence $h = 1$ in $G = M/N$. Since h is arbitrary, G is an R^* -group.

Now we verify that G will contain a nonnormal convex subgroup once a total order has been established on G . Let $1 \neq a \in G'$ and $H \prec H^\sharp$ be a convex jump determined by a . If H is normal then G/H acts on $H^\sharp/H$ by multiplication by positive real numbers, and so $H = (aH)^{x+(y-1)^2} = (\alpha_x + (\alpha_y - 1)^2)aH \neq H$, a contradiction. $\square$

We now point out solvable groups with finitely many relatively convex subgroups. Example 2.2 gives a series of orderable metabelian groups with exactly n proper relatively convex subgroups for every positive integer n . Of course, all relatively convex subgroups are normal in such groups.

Example 2.2. Let V be a direct sum of n copies of additive groups of rationals. We think of V as a vector space of dimension n over $\mathbb{Q}$ with basis $e_1, \dots, e_n$. Let t be a linear transformation of V defined thus:

$$e_1 \cdot t, \quad e_i \cdot t + e_{i-1}, \quad i = 2, \dots, n. \quad (5)$$

Since t acts on V as a Jordan block, V contains exactly $n + 1$ invariant subspaces $V_0 = \{0\}, \dots, V_i = \langle e_1, \dots, e_i \rangle, \dots, V_n = V$. Let G_n be a split product of V and $\langle t \rangle$ where the action of $\langle t \rangle$ on V is given by (5). Then G_n is an orderable metabelian group of finite rank, and hence all relatively convex subgroups of G_n are normal. It easy to see that all subgroups V_i , $i = 0, \dots, n$, are relatively convex. Relations (5) yield $[e_1, t] = e_1$, $[e_i, t] = e_i + e_{i-1}$, $i = 2, \dots, n$, and so $e_i \ll t$ for all i . Consequently, V is an absolutely convex subgroup. Moreover, all other relatively convex subgroups are contained in V because G_n/V has rank 1. If $H \leq V$ is an isolated normal subgroup of G_n then H is an invariant subspace of V . Therefore, G has only n proper relatively convex subgroups $V_1, \dots, V_n$. We also note that all V_i are absolutely convex.

3. AN ORDERED SOLVABLE GROUP WITH PERIODIC ABELIAN QUOTIENTS

In [2, Question 5.25], it was conjectured that the quotient group G/G' of a totally ordered solvable group G is not periodic. Example 3.1 gives a negative answer to this question. A minor generalization of this example allows us to construct an orderable solvable group of length four with only one proper normal relatively convex subgroup (see Example 3.4).

Example 3.1. We use an example of a right orderable group that is a generalization of Gorchakov's [7] (see also [4, Chap. I, Sec. 2, Example 9]).

Let G be a group with generators c_n , $n \in \mathbb{Z}$, and relations

$$c_k^{c_n} = \begin{cases} c_k^{-1} & \text{if } n = k + 1, \\ c_k & \text{if } n > k + 1, \end{cases} \quad (6)$$

where $n, k \in \mathbb{Z}$, $n \geq k + 1$. It follows from (6) that squares of elements of G commute, and consequently G is metabelian. Relations (6) allow every (nontrivial) element $g \in G$ to be represented as

$$g = c_{i_1}^{m_1} \dots c_{i_n}^{m_n}, \quad m_i \neq 0, \quad i_1 < \dots < i_n.$$

We call m_n the *major exponent* of an element g and write $\text{me}(g) = m_n$. Note that $\text{me}(g)$ is uniquely determined for every $1 \neq g \in G$.

Now we define subgroups $L_n = \langle c_i \mid i < n \rangle < G$ and $R_n = \langle c_i \mid i \geq n \rangle < G$, $n \in \mathbb{Z}$. The defining relations in (6) show that subgroups L_n (R_n , resp.), $n \in \mathbb{Z}$, are all isomorphic to each other and have the following properties:

LEMMA 3.2. For every integer n ,

- (1) L_n is normal in G ;
- (2) L_{n+1}/L_n is infinite cyclic;
- (3) $G = L_n R_n$;
- (4) $L_n \cap R_n = \{1\}$;
- (5) $G/L_n \cong R_n$.

The **proof** follows from the definitions. $\square$

By Lemma 3.2, mappings like

$$c_i^{\rho_n} = \begin{cases} 1 & \text{if } i < n, \\ c_i & \text{otherwise} \end{cases} \quad (7)$$

define endomorphisms $\rho_n : G \rightarrow R_n < G$ with kernels $\ker \rho_n = L_n$, $n \in \mathbb{Z}$. This gives unique decompositions such as

$$g = \lambda_n(g) \rho_n(g), \quad \lambda_n(g) \in L_n, \quad \rho_n(g) \in R_n, \quad (8)$$

for any $g, h \in G$ and any $n \in \mathbb{Z}$, where $\lambda_n(g) = g \rho_n(g)^{-1} \in L_n$.

We put $g > 1$ in G if $\text{me}(g) > 0$. This turns G into a right-ordered group with a normal series of convex subgroups of the form

$$\{1\} = \bigcap_{n \in \mathbb{Z}} L_n < \dots < L_1 < L_2 < \dots < \bigcup_{n \in \mathbb{Z}} L_n = G.$$

Now we construct a totally ordered group H as an extension of an Abelian group A by G . (Here we employ the same approach as is used in Smirnov's theorem [8]; see also [4, Chap. V, Sec. 4, Thm. 3 or 1, Thm. 3.5.1].) Smirnov's construction, however, needs to be modified since the fact that F/N has only periodic Abelian quotients does not imply that $F/[N, N]$ has the same property. To obtain an ordered group H , therefore, a special action of G on an Abelian group A will be defined throughout (9)-(14). Although such an action looks rather complicated, still, it is then easy to show that H is totally ordered and H/H' is periodic.

Let A be the free Abelian group with a basis

$$a_i^g, \quad g \in R_{i+1}, \quad i \in \mathbb{Z}, \quad b_{i,j}^f, \quad f \in R_{i+2}, \quad i \in \mathbb{Z}, \quad j \in \mathbb{N},$$

where R_i are subgroups of G defined as above. The action of G on A is defined on generators thus:

$$(a_i^g)^{c_k^{\varepsilon_k}} = a_i^{g\rho_{i+1}(c_k^{\varepsilon_k})}, \quad \varepsilon_k = \pm 1, \quad (9)$$

$$(b_{i,j}^f)^{c_k^{\varepsilon_k}} b_{i,j}^{f\rho_{i+2}(c_k^{\varepsilon_k})}, \quad \varepsilon_k = \pm 1, \quad i, k \in \mathbb{Z}, \quad j \in \mathbb{N}, \quad k \neq i+1. \quad (10)$$

To define the action of c_{i+1} on $b_{i,j}^f$, we recall that $c_{i+1}^f = c_{i+1}^{\pm 1}$ for any $f \in R_{i+2}$. Therefore, for all $i \in \mathbb{Z}$ and all $j \in \mathbb{N}$, we set

$$(b_{i,1}^f)^{c_{i+1}} = b_{i,1}^f a_i^{\varepsilon f} a_i^{\varepsilon c_{i+1}^f}, \quad (11)$$

$$(b_{i,j}^f)^{c_{i+1}} = b_{i,j}^f b_{i,j-1}^{\varepsilon c_{i+1}^f}, \quad j > 1, \quad (12)$$

where $\varepsilon = \pm 1$ is such that $c_{i+1}^f = c_{i+1}^\varepsilon$. Conjugating both sides of (11) and (12) by c_{i+1}^{-1} , we obtain

$$(b_{i,1}^f)^{c_{i+1}^{-1}} = b_{i,1}^f a_i^{-\varepsilon f} a_i^{-\varepsilon c_{i+1}^{-1} f}, \quad (13)$$

$$(b_{i,j}^f)^{c_{i+1}^{-1}} = b_{i,j}^f b_{i,j-1}^{-\varepsilon c_{i+1}^{-1} f}, \quad j > 1, \quad c_{i+1}^f = c_{i+1}^\varepsilon. \quad (14)$$

Now we need to verify that formulas (9)-(12) are consistent with defining relations for G . Relations (6) are consistent with (9) because ρ_n is an endomorphism of G for every $n \in \mathbb{Z}$. We verify the consistency of (6) with (10)-(12). Assume $c_{i+1}^f = c_{i+1}$. In this case $c_{i+1}^{f c_{i+2}^{-1}} = c_{i+1}^{-1}$. Thus,

$$\begin{aligned} (b_{i,1}^f)^{c_{i+2}^{-1} c_{i+1} c_{i+2}} &= \left(b_{i,1}^{f c_{i+2}^{-1}} \right)^{c_{i+1} c_{i+2}} \left(b_{i,1}^{f c_{i+2}^{-1}} a_i^{-f c_{i+2}^{-1}} a_i^{-c_{i+1}^{-1} f c_{i+2}^{-1}} \right)^{c_{i+2}} \\ &= \left(b_{i,1}^f a_i^{-f} a_i^{-c_{i+1}^{-1} f} \right) = (b_{i,1}^f)^{c_{i+1}^{-1}}. \end{aligned}$$

If $c_{i+1}^f = c_{i+1}^{-1}$, then $c_{i+1}^{f c_{i+2}^{-1}} = c_{i+1}$, whence

$$\begin{aligned} (b_{i,1}^f)^{c_{i+2}^{-1} c_{i+1} c_{i+2}} &= \left(b_{i,1}^{f c_{i+2}^{-1}} \right)^{c_{i+1} c_{i+2}} \left(b_{i,1}^{f c_{i+2}^{-1}} a_i^{f c_{i+2}^{-1}} a_i^{c_{i+1} f c_{i+2}^{-1}} \right)^{c_{i+2}} \\ &= (b_{i,1}^f a_i^f a_i^{c_{i+1} f}) = (b_{i,1}^f)^{c_{i+1}}. \end{aligned}$$

The consistency of defining relations with (12)-(14) can be verified in a similar way.

The split product $A \rtimes G$ of an Abelian group A and a metabelian group G defined via automorphisms (9)-(14) will be a solvable group of length three. Let

$$H = \langle c_i a_i, b_{i,j} \in A \rtimes G \mid i \in \mathbb{Z}, j \in \mathbb{N} \rangle < A \rtimes G.$$

We argue to show that H is orderable. To this end, we construct in H a series of normal subgroups satisfying conditions of Kokorin's theorem [9] (see also [4, Chap. II, Sec. 2, Thm. 3]):

A group G is orderable if and only if there exists an infrainvariant series of subgroups Σ in G satisfying the following condition: if $A \prec B$ is a jump in Σ then the quotient group B/A is $N(A)/A$ -orderable.

Put $h_i = c_i a_i$ and $H_k = \langle h_i, b_{i,j} \mid i \in \mathbb{Z}, j \in \mathbb{N}, i < k \rangle^H$, $k \in \mathbb{Z}$. The subgroups H_k form a normal series of subgroups, and $\bigcap_{k \in \mathbb{Z}} H_k = \{1\}$ and $\bigcup_{k \in \mathbb{Z}} H_k = H$ by construction. It remains to show that quotients H_{k+1}/H_k are H/H_k -orderable for every $k \in \mathbb{Z}$. Consider H_2/H_1 . By formulas (9)-(14), H_2/H_1 is a free Abelian group with basis $h_1^g, b_{1,j}^f$, $g \in R_2$, $f \in R_3$, $j \in \mathbb{N}$. Using the existing right order on G , we order this basis by setting

$$\begin{aligned}
h_1^{g_1} &\ll h_1^{g_2} \text{ if } g_1 < g_2 \text{ in } R_2; \\
h_1^g &\ll b_{1,j}^f \text{ for all } g \in R_2, f \in R_3, \text{ and for any natural } j; \\
b_{1,j}^{f_1} &\ll b_{1,k}^{f_2} \text{ if } j < k \text{ or } (j = k \text{ and } f_1 < f_2) \text{ in } R_3.
\end{aligned}$$

Then we extend the order on the basis up to a total order on H_2/H_1 . Since G is right-ordered, the so defined order on H_2/H_1 is invariant under conjugation by elements of H/H_1 (note that actions of $h_i \in H$ and $c_i \in G$ coincide on H_2). To prove that H_{k+1}/H_k is H/H_k -ordered, we need only observe that groups A , G , and H admit an automorphism

$$\delta : a_i \mapsto a_{i+1}, \quad b_{i,j} \mapsto b_{i+1,j}, \quad c_i \mapsto c_{i+1}, \quad h_i \mapsto h_{i+1}, \quad i, j \in \mathbb{Z}, \quad (15)$$

which produces an isomorphism between pairs $(H_{k+1}/H_k, H/H_k)$ and $(H_2/H_1, H/H_1)$.

Obviously, H/H' has exponent two. In fact, every one of $b_{i,j}$ is a commutator, i.e. $b_{i,j} = [b_{i,j+1}, h_{i+1}]$ in view of (12), and

$$h_i h_i^{h_{i+1}} = c_i a_i a_{i+1}^{-1} c_{i+1}^{-1} c_i a_i c_{i+1} a_{i+1} c_i a_i c_i^{-1} a_i^{c_{i+1}} = a_i a_i^{c_{i+1}}, \quad (16)$$

$$[b_{i,1}, h_{i+1}] = a_i^{-1} a_i^{-c_{i+1}} \quad (17)$$

in virtue of (10), yielding

$$h_i^2 = [h_{i+1}, h_i][b_{i,1}, h_{i+1}].$$

Hence, $g^2 \in H'$ for every $g \in H$.

Remark 3.3. In the previous example, all jumps in the series of normal convex subgroups of H are Abelian by construction. In this connection, it is worth observing that H cannot have non-Abelian jumps under any total ordering of H . This follows from a result by Medvedev [10, Thm. 2.2] (since H is metabelian-by-periodic), or from a result of Glass and Medvedev [11, Prop. 1.3] (for H belongs to a variety $\mathcal{AU}$, where $\mathcal{A}$ is a variety of Abelian groups and $\mathcal{U}$ is a variety defined by an unilateral identity $[x, y^2, y^2] = 1$). We also note that all orderable groups in $\mathcal{U}$ are nilpotent, but H/H'' is not.

Example 3.4. Now we use the group G given in Example 3.1 to construct a new group K as a split product of G and an infinite cyclic group $\langle d \rangle$, where d acts on H by conjugation as does the automorphism δ defined in (15). The group K is solvable of length at most four (as an extension of a length three solvable group by an Abelian group). In fact, the derived length of this group equals four because the third derived subgroup of K contains nontrivial elements. One such element will be needed below, and we so calculate it now.

First, $b_{1,4} = [b_{1,5}, h_2]$ is a commutator. Hence,

$$b_{1,3} = [b_{1,4}, h_2] = [b_{1,4}, [h_1, d]]$$

is a commutator of derived length two. Furthermore,

$$u_{1,1} = [h_1, d^2] = h_1^{-1} h_3 \quad \text{and} \quad u_{1,2} = [h_2, d] = h_2^{-1} h_3 \quad (18)$$

are also commutators, and so

$$u_2 = [u_{1,1}, u_{1,2}] = h_3^{-1} h_1 h_3^{-1} h_2 h_1^{-1} h_3 h_2^{-1} h_3 c_2^2 c_1^2 a_2^{c_3^2 - c_3} a_1^{c_2^{-2} c_3 - c_2 c_3^2} \quad (19)$$

is a commutator of derived length two. Consequently,

$$[b_{1,3}, u_2] = b_{1,3}^{-1} b_{1,3}^{c_2^2} = b_{1,1} b_{1,2}^2 \quad (20)$$

is a commutator of derived length three. Since $b_{i,k}$ form a basis for a free Abelian group B , it follows that $1 \neq b_{1,1}b_{1,2}^2 \in K'''$ and the derived length of K equals four.

Let N_α , $\alpha \in I$, be a chain of normal relatively convex subgroups of K with $b_{1,1}b_{1,2}^2 \notin N_\alpha$. Then $N = \bigcup_{\alpha \in I} N_\alpha$ is also a normal relatively convex subgroup of K , and $b_{1,1}b_{1,2}^2 \notin N$. In view of (18)-(20), $d \notin N$. Consider $\bar{K} = K/N$. Then:

- (1) $\bar{b}_{1,1}\bar{b}_{1,2}^2 \neq 1$ in $\bar{K}$, and so $\bar{K}$ is solvable of length four.
- (2) $\bar{K}$ is orderable, and if T is a normal relatively convex subgroup of $\bar{K}$, then $\bar{b}_{1,1}\bar{b}_{1,2}^2 \in T$. Hence, $\bar{b}_{1,2} \in T$ because $\bar{b}_{1,1}\bar{b}_{1,2}^2 = \bar{b}_{1,2}\bar{b}_{1,2}^{\bar{h}_2}$ and T is strictly isolated. Thus, $\bar{b}_{1,1} \in T$.

In view of $\bar{K}/\bar{H} \cong \langle d \rangle$, the subgroup $\bar{H}$ is relatively convex. We show that $\bar{K}$ has no other proper normal relatively convex subgroups. Assume that T is a proper normal relatively convex subgroup of $\bar{K}$.

As shown in (2), $\bar{b}_{1,1} \in T$. Since T is normal, $[\bar{h}_2, \bar{b}_{1,1}] = \bar{a}_1\bar{a}_1^{\bar{c}_2} \in T$, and in view of (16), $\bar{h}_1\bar{h}_1^{\bar{h}_2} \in T$. Since T is strictly isolated, $\bar{h}_1 \in T$. This gives $\bar{h}_i = \bar{h}_1^{\bar{q}_i-1} \in T$ for any $i \in \mathbb{Z}$. Finally, $\bar{b}_{i,j} = [\bar{b}_{i,j+1}, \bar{h}_{i+1}] \in T$ for all $i, j \in \mathbb{Z}$. Hence, $\bar{H} \leq T$, and so $\bar{K}/T$ is relatively convex in $\bar{K}/\bar{H}$, whence $\bar{H} = T$.

Remark 3.5. The group $\bar{K}$ in Example 3.4 was not finitely generated because the number of elements $\bar{b}_{i,j}$ was infinite; but we used only one such element, namely, $\bar{b}_{1,5}$. Thus, a subgroup $\bar{K}_1 = \langle \bar{d}, \bar{h}_1, \bar{b}_{1,5} \rangle < \bar{K}$ is 3-generated and has the same property as has $\bar{K}$, i.e., $\bar{K}_1$ is solvable of length four with a sole proper normal relatively convex subgroup. We are therefore faced up to an example of a finitely generated solvable group of length four with only one proper normal relatively convex subgroup.

REFERENCES

1. V. M. Kopytov and N. Ya. Medvedev, *Right Ordered Groups* [in Russian], Nauch. Kniga, Novosibirsk (1996).
2. *Unsolved Problems in Group Theory, The Kourovka Notebook*, 16th edn., Institute of Mathematics SO RAN, Novosibirsk (2006); <http://www.math.nsc.ru/~alglog>.
3. A. Rhemtulla and H. Smith, "On solvable R^* groups of finite rank," *Com. Alg.*, **31**, No. 7, 3287–3293 (2003).
4. A. I. Kokorin and V. M. Kopytov, *Linearly Ordered Groups* [in Russian], Nauka, Moscow (1972).
5. R. B. Mura and A. H. Rhemtulla, *Orderable Groups, Lect. Notes Pure Appl. Math.*, **27**, Marcel Dekker, New York (1977).
6. P. H. Kropholler, "On finitely generated soluble groups with no large wreath product sections," *Proc. London Math. Soc.*, *III. Ser.*, **49**, No. 1, 155–169 (1984).
7. Yu. M. Gorchakov, "An example of G -periodical torsion-free group," *Algebra Logika*, **6**, No. 3, 5–7 (1967).
8. D. M. Smirnov, "Right ordered groups," *Algebra Logika*, **5**, No. 6, 41–59 (1966).
9. A. I. Kokorin, "T- O^* -subgroups and relatively convex subgroups of ordered groups," *Sib. Mat. Zh.*, **7**, No. 3, 713–717 (1966).
10. N. Ya. Medvedev, "Soluble groups and varieties of ℓ -groups," *Algebra Logika*, **44**, No. 3, 355–367 (2005).
11. A. M. Glass and N. Ya. Medvedev, "Unilateral o -groups," *Algebra Logika*, **45**, No. 1, 20–27 (2006).